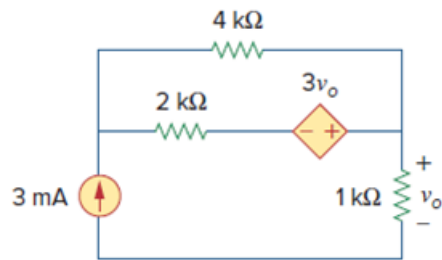


Problem

Use source transformation to find v_o in the circuit of Fig. 4.97.

Figure 4.97:

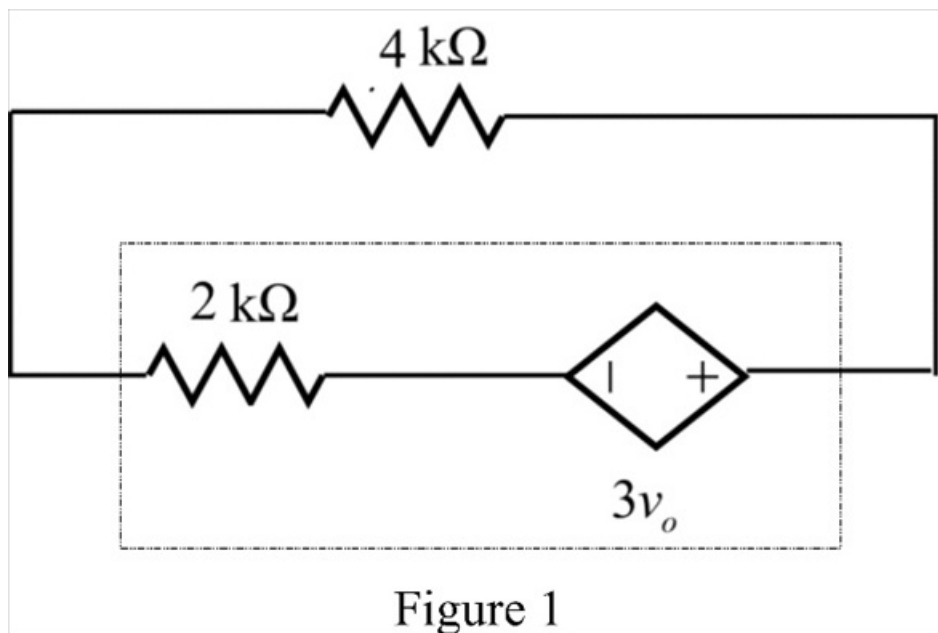


Step-by-step solution

Step 1 of 5

Refer to Figure 4.97 in the textbook.

Consider the following part from the Figure 4.97 as shown in Figure 1.



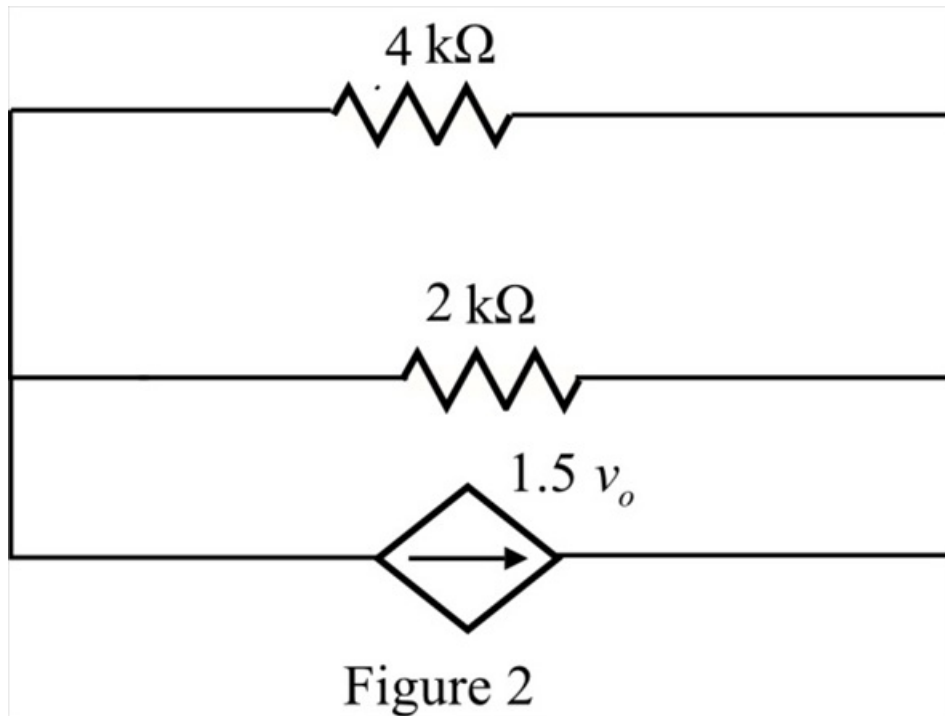
Step 2 of 5

Determine the equivalent circuit to Figure 1.

$$I = \frac{3v_o}{2}$$

$$= 1.5v_o \text{ A}$$

Draw the circuit as shown in Figure 2.



Step 3 of 5

Replace Figure 2 with its equivalent circuit.

Consider $4 \text{ k}\Omega$ and $2 \text{ k}\Omega$ are in parallel combination.

$$R_{eq} = 4 \parallel 2$$

$$= \frac{8}{6}$$

$$= \frac{4}{3} \Omega$$

Now change the current source into the voltage source.

Determine the equivalent voltage source.

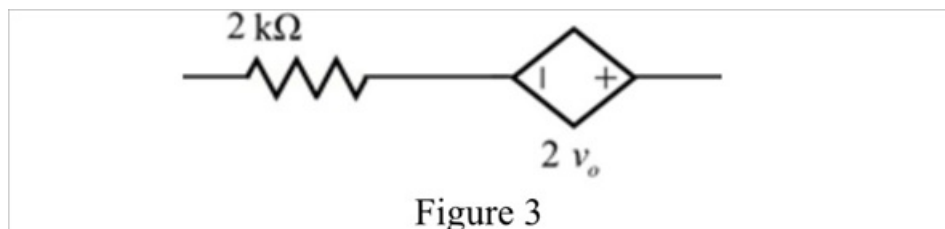
$$v = (1.5 v_o) R_{eq}$$

$$= (1.5 v_o) \left(\frac{4}{3} \right)$$

$$= 2 v_o$$

Step 4 of 5

Draw the figure for this part only.



Draw the circuit as shown in Figure 4.

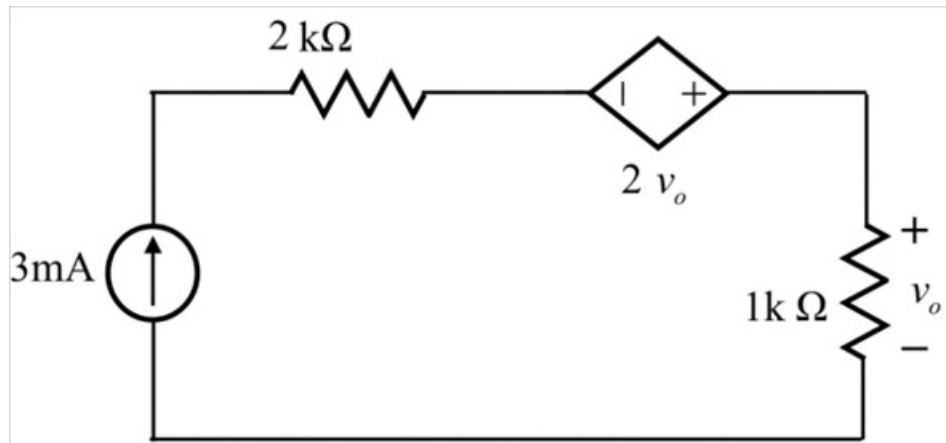


Figure 4

Determine the voltage across 1 kΩ.

$$\begin{aligned} v_o &= (3 \text{ mA})(1 \text{ k}\Omega) \\ &= 3 \text{ V} \end{aligned}$$

Therefore, the value of the v_o is 3 V.